

Project Design Phase-II Technology Stack (Architecture & Stack)

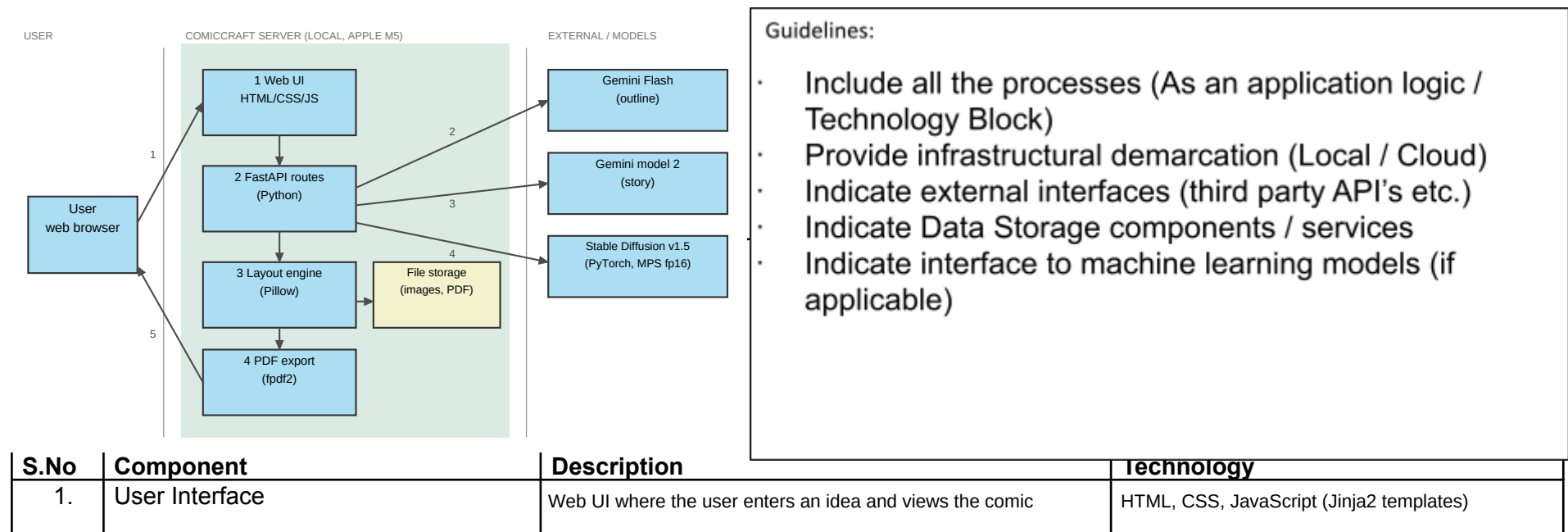
Date	2 October 2026
Team ID	12
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: AI comic story creation (ComicCraft)

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



2.	Application Logic-1	FastAPI routes receive requests and orchestrate the pipeline	Python, FastAPI, Uvicorn
3.	Application Logic-2	Outline and story generation with Gemini	Google Gemini Flash and a second Gemini model
4.	Application Logic-3	Panel layout, caption fitting and PDF export	Python, Pillow, fpdf2
5.	Database	No database; each request writes plain files	Local filesystem
6.	Cloud Database	Not used (stateless application)	N/A
7.	File Storage	Generated panel images and exported PDFs	Local filesystem
8.	External API-1	Story outline generation (Gemini Flash)	Google Gemini API (free tier)
9.	External API-2	Full story script (second Gemini model)	Google Gemini API (free tier)
10.	Machine Learning Model	Text-to-image generation for each comic panel	Stable Diffusion v1.5 (diffusers, PyTorch)
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Apple M5, 16 GB RAM, MPS fp16 Cloud Server Configuration: Not used	Local (Uvicorn on Apple Silicon)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks and libraries used by the application	FastAPI, Uvicorn, PyTorch, diffusers, Pillow, fpdf2, pytest
2.	Security Implementations	Secrets in environment file; no user data is stored	API keys in environment variables (.env); server-side input validation
3.	Scalable Architecture	Layered modules (UI, routes, AI adapters, layout, export); can move to a queue and cloud GPU	Stateless routes; models can move to a GPU server or cloud endpoint
4.	Availability	Local server; Gemini quota errors are caught, retried and reported	Runs locally; retries on 429 and a second Gemini model for quota

S.No	Characteristics	Description	Technology
5.	Performance	About 12.4 s per panel image and 104 s end to end; 24 automated tests guard regressions	Stable Diffusion fp16 on MPS; pipeline loaded once and reused

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>